

On-Demand Resources (ODR) in iOS

Download only what users need, when they need it.



ASSETS STREAM FROM THE APP STORE — ON DEMAND



Smaller App Size

Reduce initial
download size



Better Experience

Users get what they
need, faster



Smarter Delivery

Optimized for storage
and performance

What is On-Demand Resources?



On-Demand Resources (ODR) allows your app to download assets only when they are needed, instead of bundling everything in the IPA.

Benefits



Smaller App Size

Lower initial download size



Faster Downloads

Users get to the core app quickly



Better User Experience

Access features on demand



Lower Storage Usage

iOS manages and removes unused resources

How On-Demand Resources **Work**



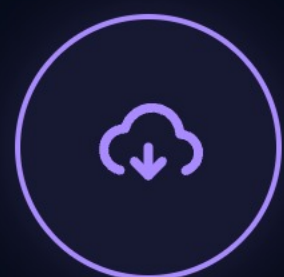
1. App Installed

Only essential resources are included.



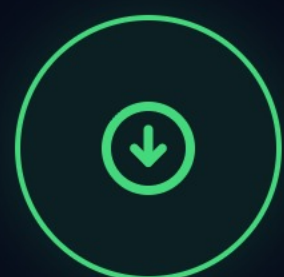
2. User Opens Feature

Feature requires additional assets.



3. Request ODR Asset

App requests the required resource from the App Store.



4. Downloaded & Used

iOS downloads the asset and makes it available.



5. System Removes Later

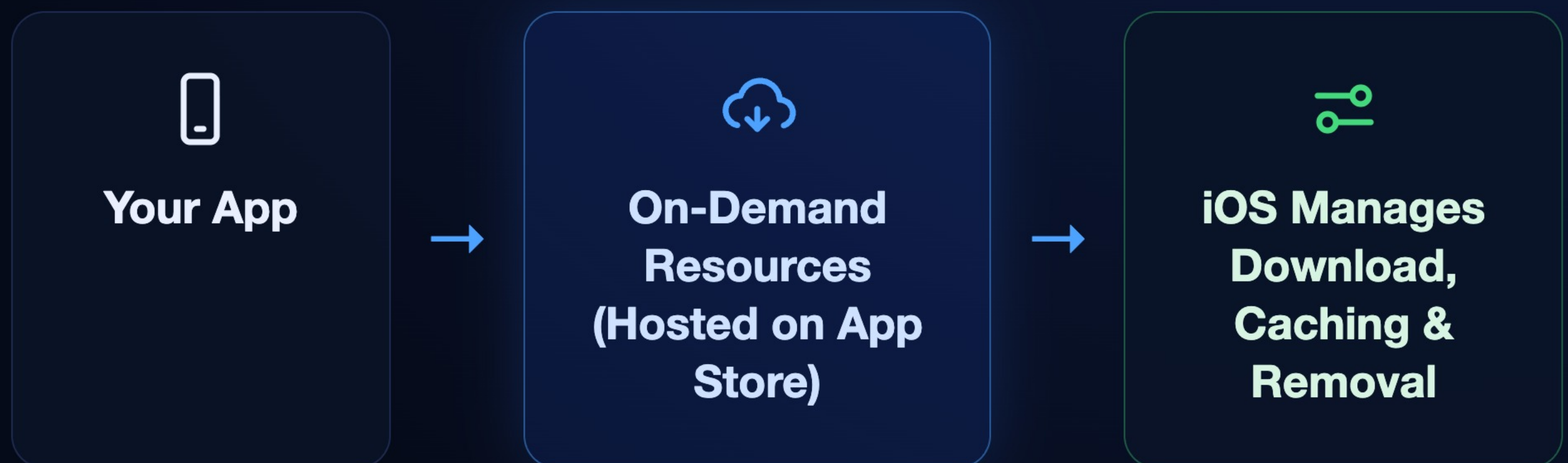
If storage is needed, iOS may remove it automatically.



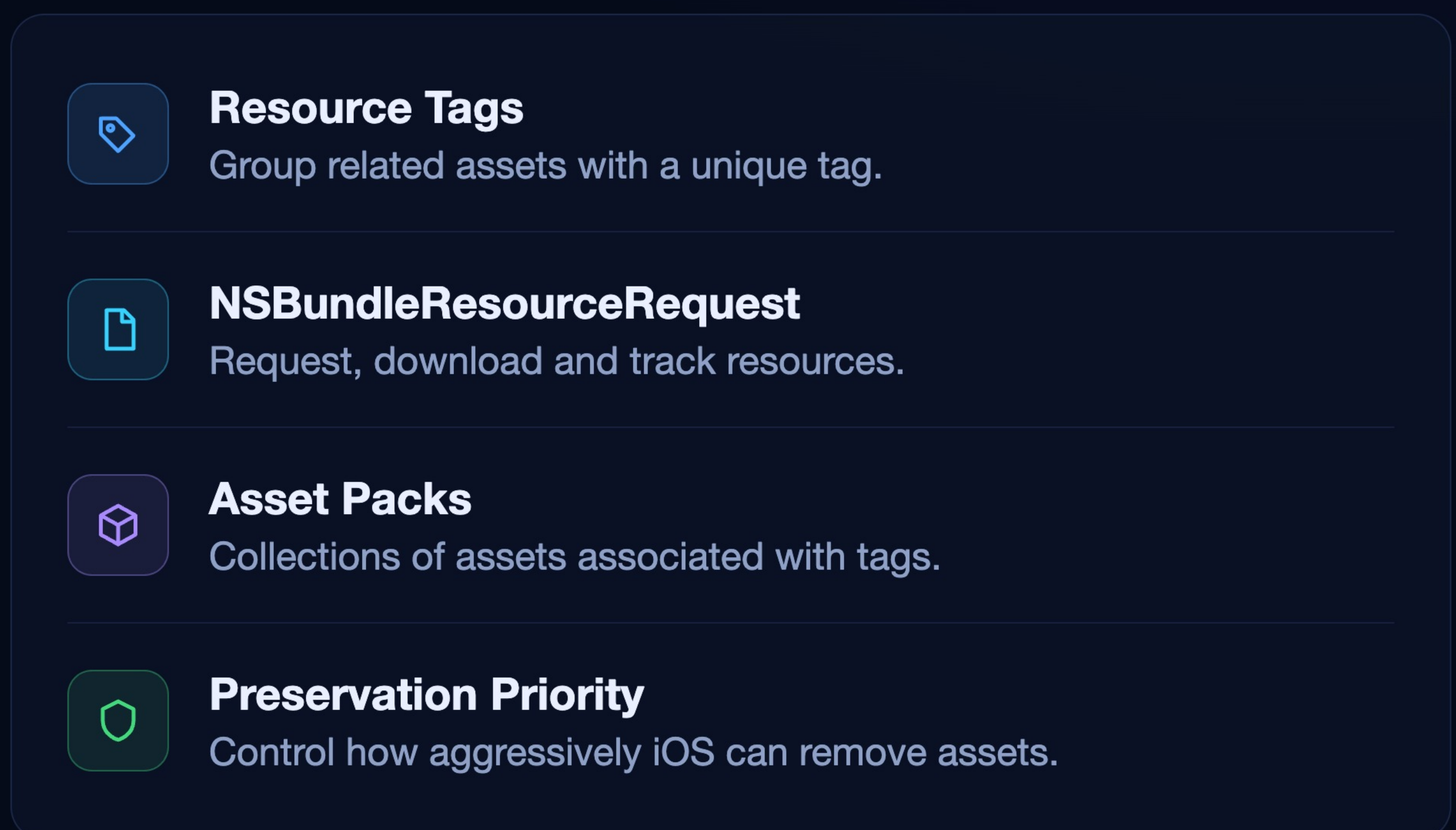
The app can download it again whenever needed.

ODR Architecture & Key Components

High-level Architecture



Key Components



When to **Use** (and **Avoid**) On-Demand Resources

✓ Use ODR For

- Game levels
- Large images / galleries
- Tutorial videos
- Offline maps
- AR / 3D assets
- Downloadable themes
- Seasonal / regional content
- PDF / document packs

✗ Don't Use ODR For

- App icons
- Launch screen
- Core UI assets
- Fonts
- Login / auth assets
- App logo
- Frequently used images
- Assets needed at launch

Rule of Thumb

- ★ If the user needs it immediately after launch, it's probably not a good candidate for ODR.

Implementing ODR in iOS (Swift)

1. Create a resource request

```
let request = NSBundleResourceRequest(tags: ["level1"])
request.beginAccessingResources { error in
    if let error = error {
        print("Failed: \(error.localizedDescription)")
    } else {
        print("Resources ready to use!")
    }
}
```

2. Monitor progress

```
let progress = request.progress
print("Progress: \(progress.fractionCompleted)")
```

3. Release when done

```
request.endAccessingResources()
```



iOS may remove these resources later to free up storage.

Best Practices, Benefits & Key Takeaways

Best Practices

- ✓ Keep core app independent of ODR.
- ✓ Group assets using meaningful tags.
- ✓ Prefetch resources before users need them.
- ✓ Show progress and handle failures gracefully.
- ✓ Release resources after use.
- ✓ Test on real devices with low storage.

Benefits Recap

- Smaller IPA size
- Better App Store conversion
- Saves device storage
- Faster updates (only new assets)
- Scalable & future-proof architecture

💬 Interview Tip

Q: Why not store everything inside the app?

A: Because most users never use every feature. ODR keeps the app lightweight and downloads assets only when needed — saving bandwidth and storage while improving user experience.

Key Takeaway

- ⬇️ Deliver the right assets, to the right users, at the right time.
That's the power of On-Demand Resources.